

Supplemental Online Content

Wijesurendra R, Pessoa-Amorim G, Buck G, et al. Remote screening for asymptomatic atrial fibrillation: the AMALFI randomized clinical trial. *JAMA*. Published online August 29, 2025. doi:10.1001/jama.2025.15440

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This supplemental material has been provided by the authors to give readers additional information about their work.

eMethods

Patch monitor

The Zio XT patch is a single lead (modified lead II ECG) applied to the chest that records up to 14 days of continuous ECG data which has been Food and Drug Administration (FDA) cleared and UK Conformity Assessed (UKCA) approved. Previous studies have shown higher diagnostic yield and lower retesting rates with Zio XT compared to shorter duration, intermittent monitoring, and other long-term continuous monitoring devices.¹⁻³ Home-based application and activation of the Zio XT has been previously used in clinical trials and in clinical practice and has been shown to have comparable wear time and analyzable time compared to in-clinic application⁴.

After the 14-day period (or before, if the patch fell off or was removed), participants were asked to mail the patch back to the manufacturer using a pre-paid return mailer. A reminder letter was sent to those who had not returned their patch after several weeks.

The recorded ECG data were analyzed by iRhythm using a validated proprietary deep-learning algorithm that is capable of detecting 13 arrhythmias (including atrial fibrillation [AF]), plus sinus rhythm and noise. The algorithm has previously shown high diagnostic performance when compared with a committee of certified cardiologists.⁵ AF was defined as the presence of at least 30s of continuous AF-compatible tracing during monitoring. ECG findings were reviewed and classified by qualified cardiovascular technicians at iRhythm,¹ and a technician report of preliminary findings was provided to the study team using a secure web-based system. No manual classification or over-read were performed by the study team. Automated alert systems were in place for AF and other arrhythmias considered by the study team to be clinically actionable (i.e., atrial flutter; sustained ventricular tachycardia or fibrillation; pauses lasting over 6s; complete heart block, Mobitz type II, or high-grade atrioventricular block), which were communicated to general practitioners (GPs) immediately by secure email. Separately, the study team received regular structured data extracts from iRhythm including all ECG findings, which were used to automatically generate results letters that were sent to GPs, with copies to participants. The letters detailed the presence or absence of AF, along with duration, heart rate, and burden. GPs managed any findings at their discretion as part of clinical care; no guidance on the management of AF or other arrhythmias was provided by AMALFI, besides referencing relevant guidelines.^{6,7}

eMethods References

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eTable 1. Patch usage metrics

	Randomized to patch (N = 2520)
Wore and returned patch - n (%)	2126 (84.4%)
Days from randomization to patch activation - Median (IQR)	10 (6 - 18)
Days wore patch - Median (IQR)	13.9 (13.2 - 14.0)
Days patch data analyzable - Median (IQR)	13.5 (12.6 - 13.9)
Percentage of patch data suitable for analysis - Median (IQR)	98.8 (95.8 - 99.6)
Patch worn ≥10.5 days (75% recommended time) - n (%)	1960 (77.8%)
Patch worn but <10.5 days - n (%)	166 (6.6%)
Reasons patch removed - n (%)*	
Patch attachment problems	43 (25.9%)
Skin reaction	18 (10.8%)
Other	8 (4.8%)
Unknown	97 (58.4%)
Patch returned unworn - n (%)	188 (7.5%)
Patch not returned - n (%)	206 (8.2%)

IQR=Interquartile range, SD=Standard deviation. * Percent of the 166 who wore the patch <10.5 days

eTable 2. Baseline characteristics by patch ECG data availability

	Overall in Patch group N = 2520	Patch ECG data N = 2126 (84.4%)	No patch ECG data N = 394 (15.6%)	P-value
Age at randomization (years) - n (%)				
<75	683	577 (84.5%)	106 (15.5%)	0.75/0.62 (trend)
≥75, <80	1032	876 (84.9%)	156 (15.1%)	
≥80	805	673 (83.6%)	132 (16.4%)	
Mean (SD)	77.7 (5.9)	77.6 (5.9)	77.8 (6.1)	0.50
Sex - n (%)				
Males	1340	1137 (84.9%)	203 (15.1%)	0.47
Females	1180	989 (83.8%)	191 (16.2%)	
Race and ethnicity ^a - n (%)				
Asian	27	20 (74.1%)	7 (25.9%)	0.17
Black	2	2 (100.0%)	0 (0.0%)	
White	2045	1738 (85.0%)	307 (15.0%)	
Other ^b	21	17 (81.0%)	4 (19.0%)	
Missing	425	349 (82.1%)	76 (17.9%)	
Body mass index (kg/m ²) - n (%)				
<25	488	422 (86.5%)	66 (13.5%)	0.07/0.03 (trend)
25 to <30	751	639 (85.1%)	112 (14.9%)	
≥30	571	466 (81.6%)	105 (18.4%)	
Missing	710	599 (84.4%)	111 (15.6%)	
Mean (SD)	28.2 (5.2)	28.0 (5.0)	29.3 (6.1)	<0.001
CHA ₂ DS ₂ VAsc score - n (%)				
<3	33	31 (93.9%)	2 (6.1%)	<0.001/<0.001 (trend)
3	685	596 (87.0%)	89 (13.0%)	
4	983	846 (86.1%)	137 (13.9%)	
≥5	819	653 (79.7%)	166 (20.3%)	
Median (IQR)	4 (3 - 5)	4 (3 - 5)	4 (4 - 5)	<0.001
Prior diagnoses at randomization - n (%)				
Hypertension	2255	1908 (84.6%)	347 (15.4%)	0.32
Diabetes	713	582 (81.6%)	131 (18.4%)	0.02
Stroke or TIA	485	395 (81.4%)	90 (18.6%)	0.05
Chronic kidney disease (stage ≥3)	438	366 (83.6%)	72 (16.4%)	0.61
Thromboembolism	271	216 (79.7%)	55 (20.3%)	0.03
Myocardial infarction	269	226 (84.0%)	43 (16.0%)	0.87
Heart failure	244	191 (78.3%)	53 (21.7%)	0.006
Peripheral arterial disease	214	170 (79.4%)	44 (20.6%)	0.04
Medication prior to randomization - n (%)				
Statin	1739	1462 (84.1%)	277 (15.9%)	0.54
RAAS inhibition	1599	1341 (83.9%)	258 (16.1%)	0.36
Proton-pump inhibitors or H2-antagonists	1265	1065 (84.2%)	200 (15.8%)	0.81

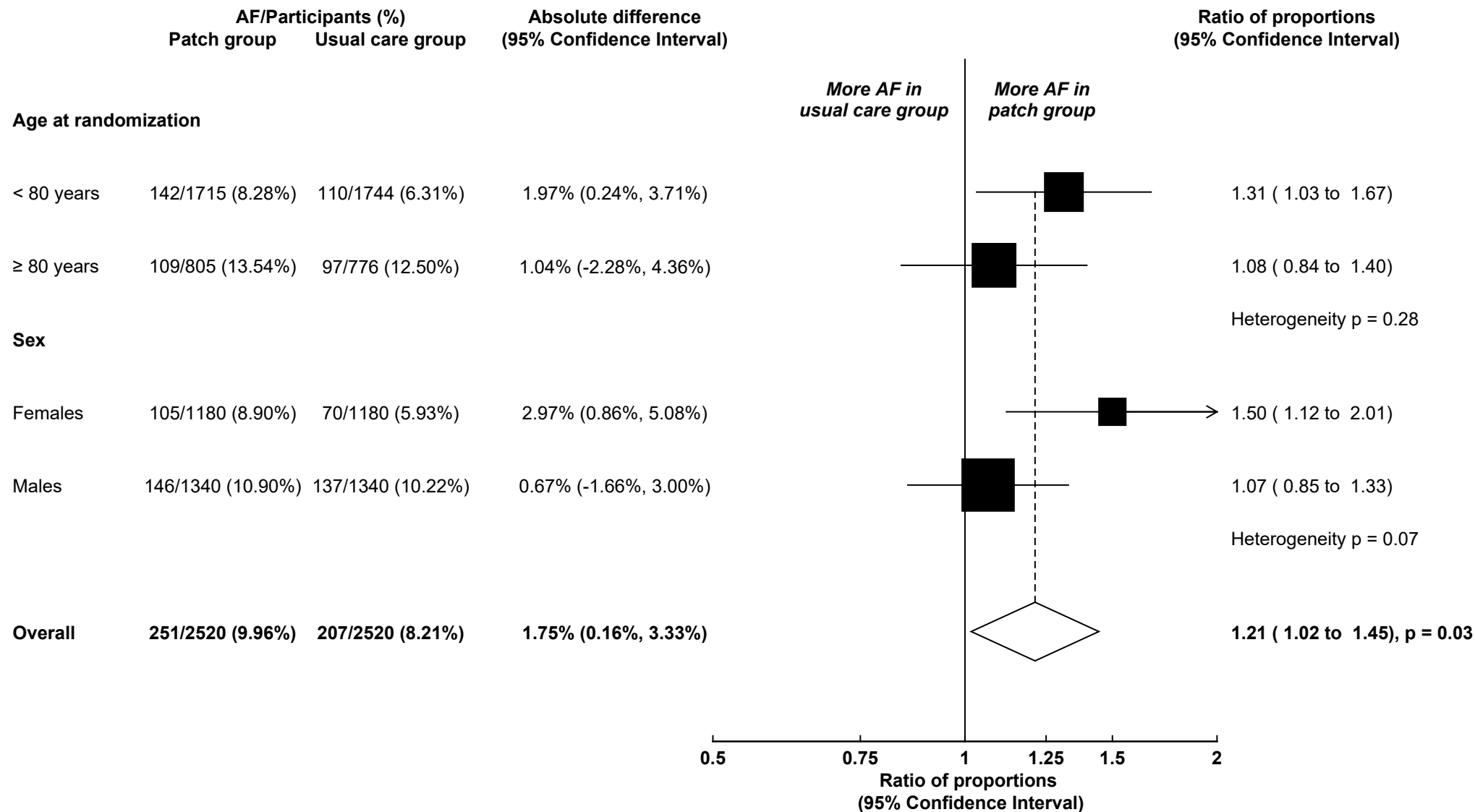
	Overall in Patch group N = 2520	Patch ECG data N = 2126 (84.4%)	No patch ECG data N = 394 (15.6%)	P-value
Calcium-channel blockers	1200	996 (83.0%)	204 (17.0%)	0.07
Diuretics	713	593 (83.2%)	120 (16.8%)	0.30
Aspirin or dipyridamole	705	585 (83.0%)	120 (17.0%)	0.23
Beta-blockers	619	529 (85.5%)	90 (14.5%)	0.39
P2Y12 inhibition	363	300 (82.6%)	63 (17.4%)	0.33
Oral anticoagulation	174	137 (78.7%)	37 (21.3%)	0.03
Direct oral anticoagulant	128	101 (78.9%)	27 (21.1%)	0.08
Vitamin K antagonist	59	47 (79.7%)	12 (20.3%)	0.31
Insulin	117	94 (80.3%)	23 (19.7%)	0.22

IQR = Interquartile range, SD = standard deviation, RAAS =Renin Angiotensin Aldosterone System. Row percentages are presented to provide the compliance rate within each category of baseline characteristic. Chi-square p-values are presented for categorical factors with the Mantel- Haenszel test for trend also presented for ordinal categorical factors. Two-sample T-test and Mann-Whitney U-test p-values are presented for continuous data with normal and non-normal distributions respectively. Missing data is excluded in the calculation of p-values. ^aP-value compares white versus any other non-missing race and ethnicity group. ^bOther category includes multiracial ethnicities (n=10) and 'any other ethnicity' (code S) recorded in secondary care (n=11).

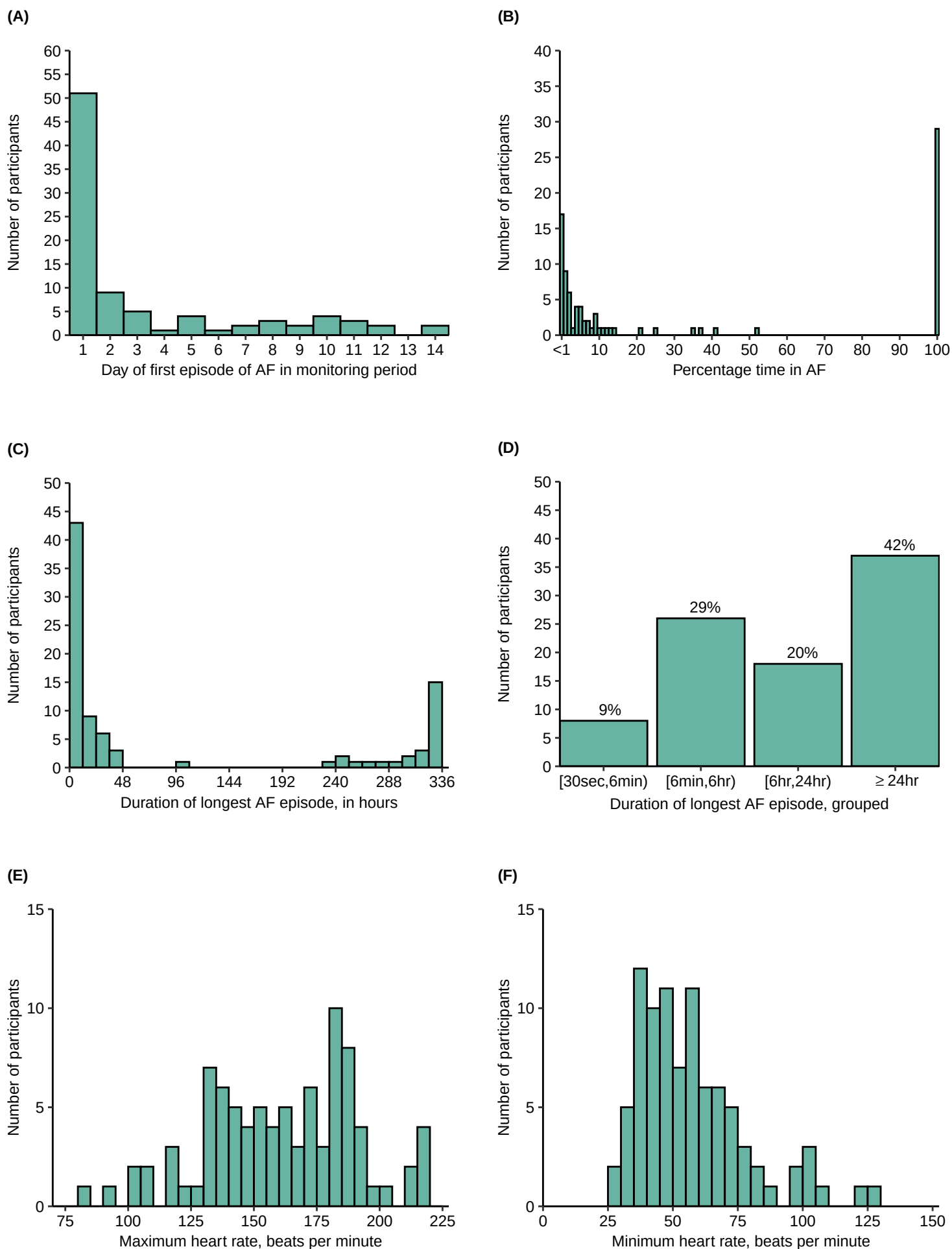
eTable 3. Patch-detected conditions

	Number of participants with condition detected	% of those randomized to patch (N = 2520)	% of those who wore and returned patch (N = 2126)
Atrial fibrillation	89	3.53%	4.19%
Atrial fibrillation without flutter	74	2.94%	3.48%
Atrial fibrillation with flutter	15	0.60%	0.71%
Other findings resulting in urgent reporting to GP			
Atrial flutter without fibrillation	13	0.52%	0.61%
Complete AV block	5	0.20%	0.24%
High grade AV block	10	0.40%	0.47%
AV block second degree Mobitz II	8	0.32%	0.38%
AV block second degree 2:1	1	0.04%	0.05%
Pauses lasting longer than 6 seconds	6	0.24%	0.28%
Sustained ventricular tachycardia (>30 seconds)	2	0.08%	0.09%
Non-urgent findings			
Non-sustained ventricular tachycardia (≤30 seconds)	676	26.83%	31.80%
Sustained supraventricular tachycardia (>30 seconds)	134	5.32%	6.30%
Non-sustained supraventricular tachycardia (≤30 seconds)	1772	70.32%	83.35%

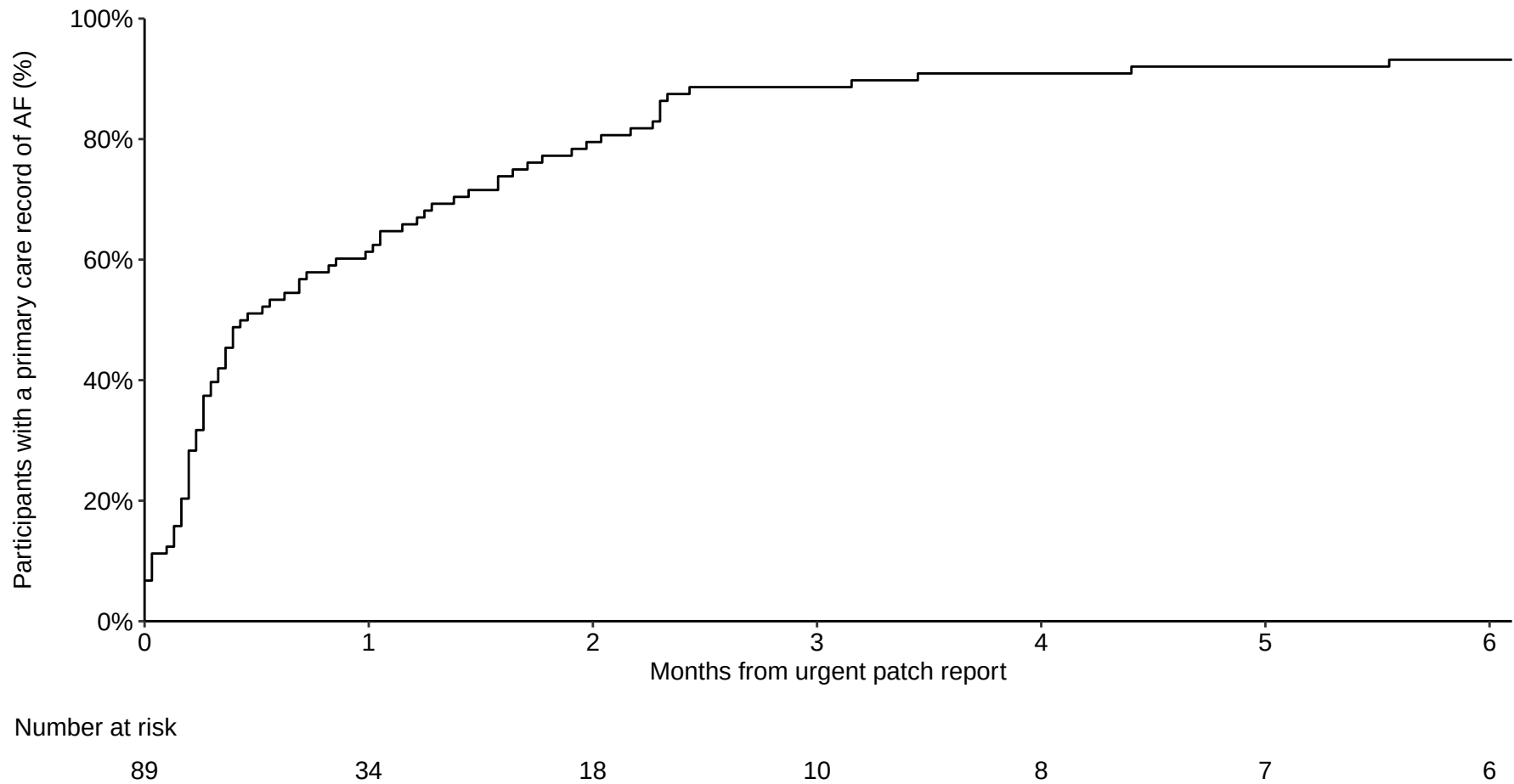
AV = Atrioventricular, GP = General practitioner



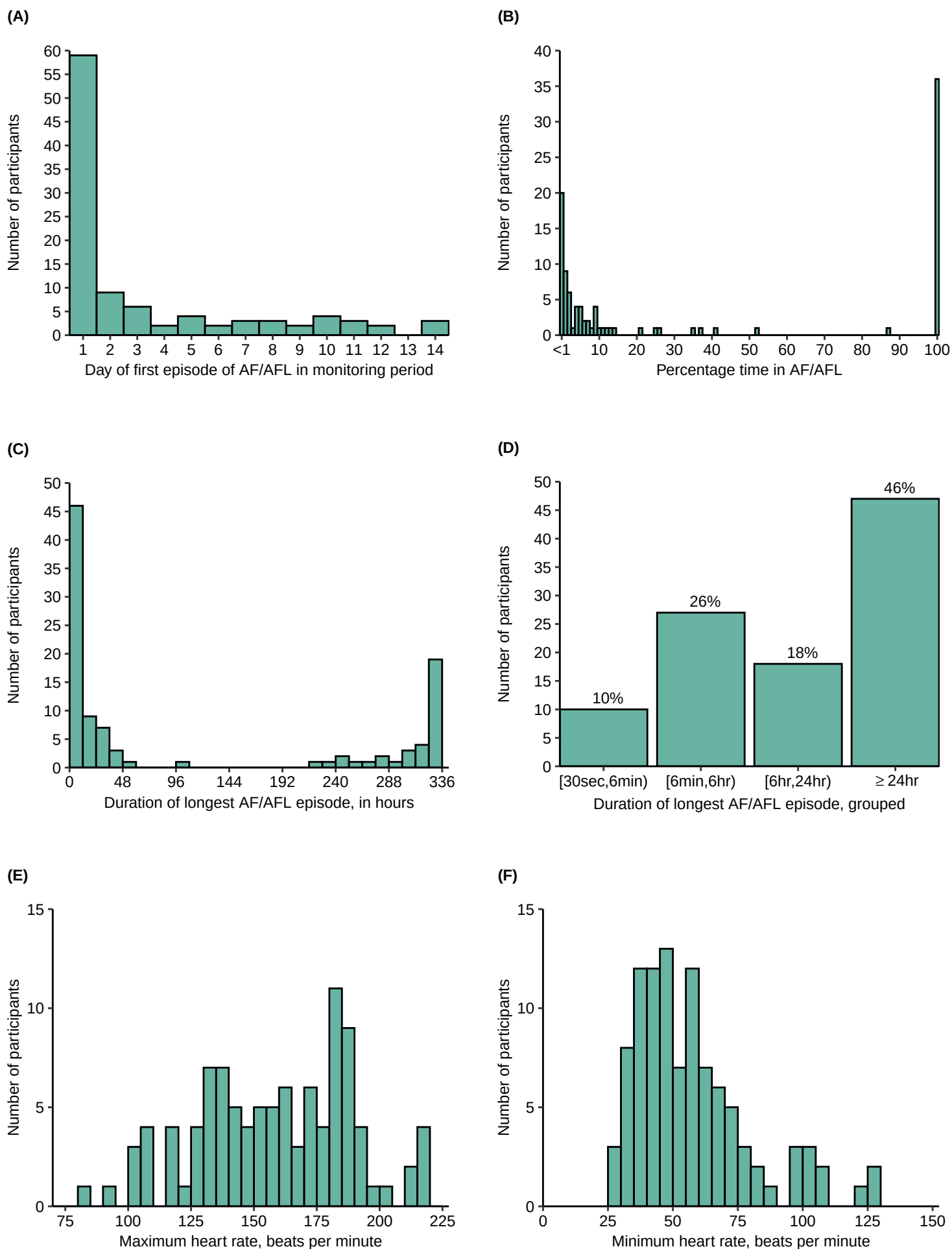
eFigure 1. Proportions, absolute differences and ratios of proportions of participants with a primary care or secondary care record of atrial fibrillation (AF) between randomization and 2.5 years by randomized allocation overall and in age and sex subgroups. Ratios of proportions in subgroups are plotted as squares with the size of each square proportional to the amount of statistical information available. The primary outcome of ratio of proportions overall is represented by a diamond. The horizontal lines and the width of the diamond represent 95% confidence intervals (not adjusted for multiplicity), and the dashed vertical line indicates the overall ratio of proportions for the effect of patch group on AF. Squares or a diamond to the right of the solid vertical line indicate a higher proportion of AF in the patch group, but the comparison is only significant (at $p < 0.05$) if the horizontal line, or diamond, does not overlap with the solid vertical line.



eFigure 2. Patch-detected atrial fibrillation metrics. (A) Day of first episode (B) Burden (C) Duration of longest episode, in hours (D) Duration of longest episode, grouped (E) Maximum heart rate during episode (F) Minimum heart rate during episode



eFigure 3. Kaplan–Meier plot of time from urgent patch report to first primary care record of atrial fibrillation (AF), restricted to those with patch–detected AF. The number of participants at risk at the start of each month is shown beneath the figure. The seven participants with a post-randomization record of AF in the primary care record that pre-dates the urgent patch report are included in the number at risk at day 0 and counted as an event immediately after this.



eFigure 4. Patch-detected atrial fibrillation and/or flutter (AF/AFL) metrics. (A) Day of first episode (B) Burden (C) Duration of longest episode, in hours (D) Duration of longest episode, grouped (E) Maximum heart rate during episode (F) Minimum heart rate during episode.